Machine Learning Course GUI

Data Management

Split Strategy: Train/Test Split | Dataset: Load Custom Dataset | Load Data | Scaling: No Scaling | Test Split: 0,20 | Scaling: | Handle Missing: Do Nothing | Split Ratio: 80-20 | K-Fold CV: 5

Classical ML | Deep Learning | Dimensionality Reduction | Reinforcement Learning | Advanced Visualization

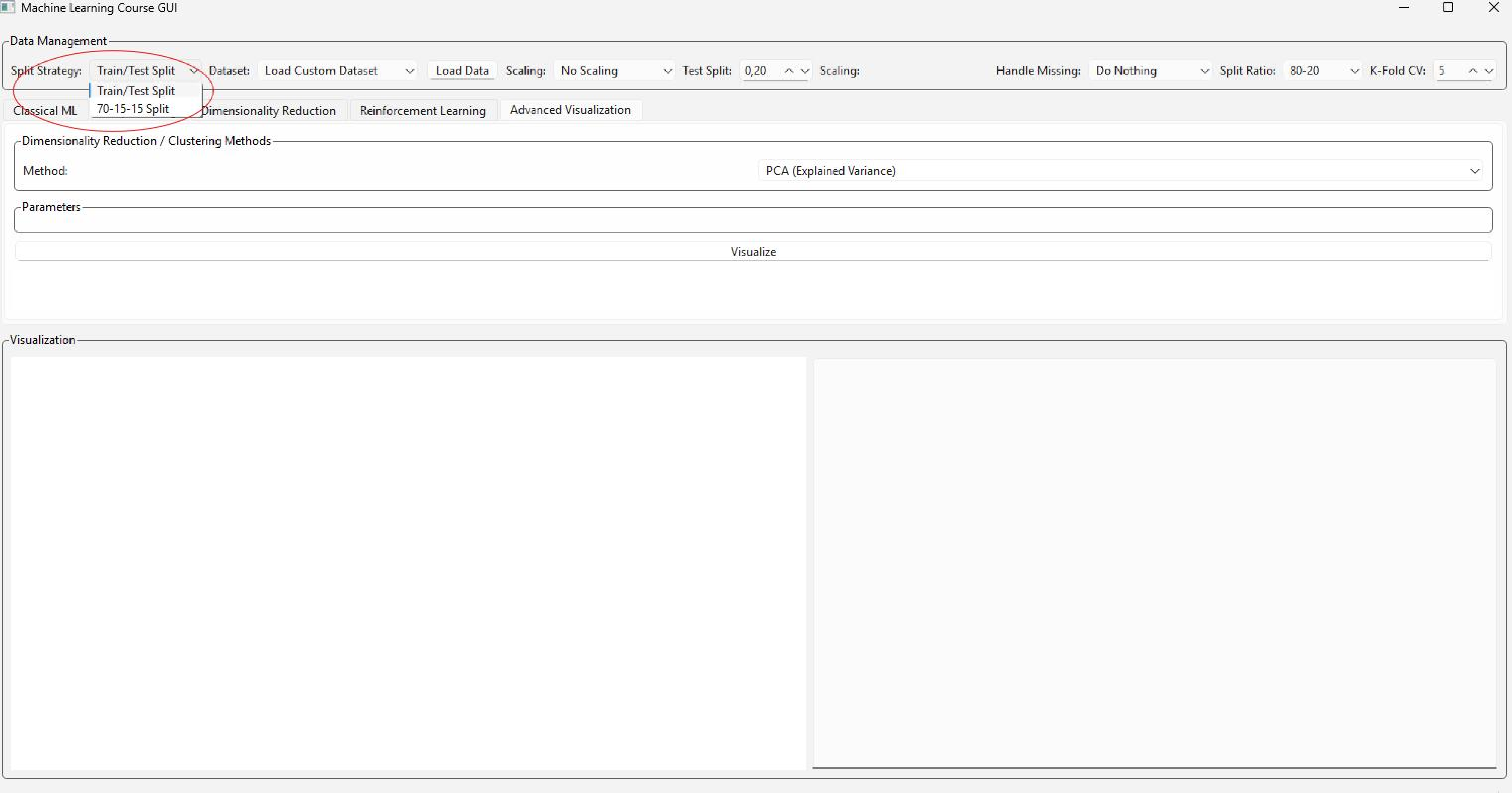
Dimensionality Reduction / Clustering Methods

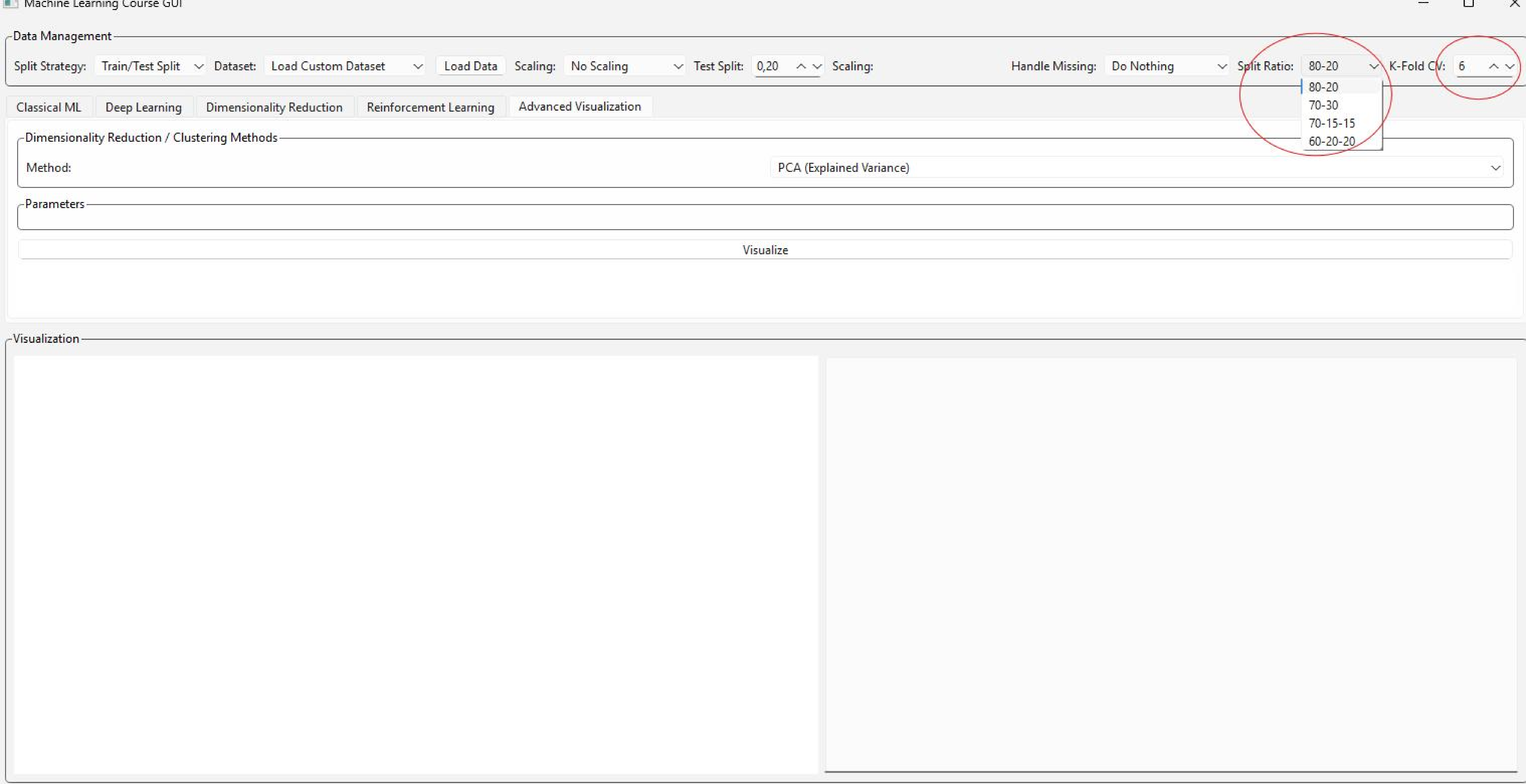
Method: PCA (Explained Variance)

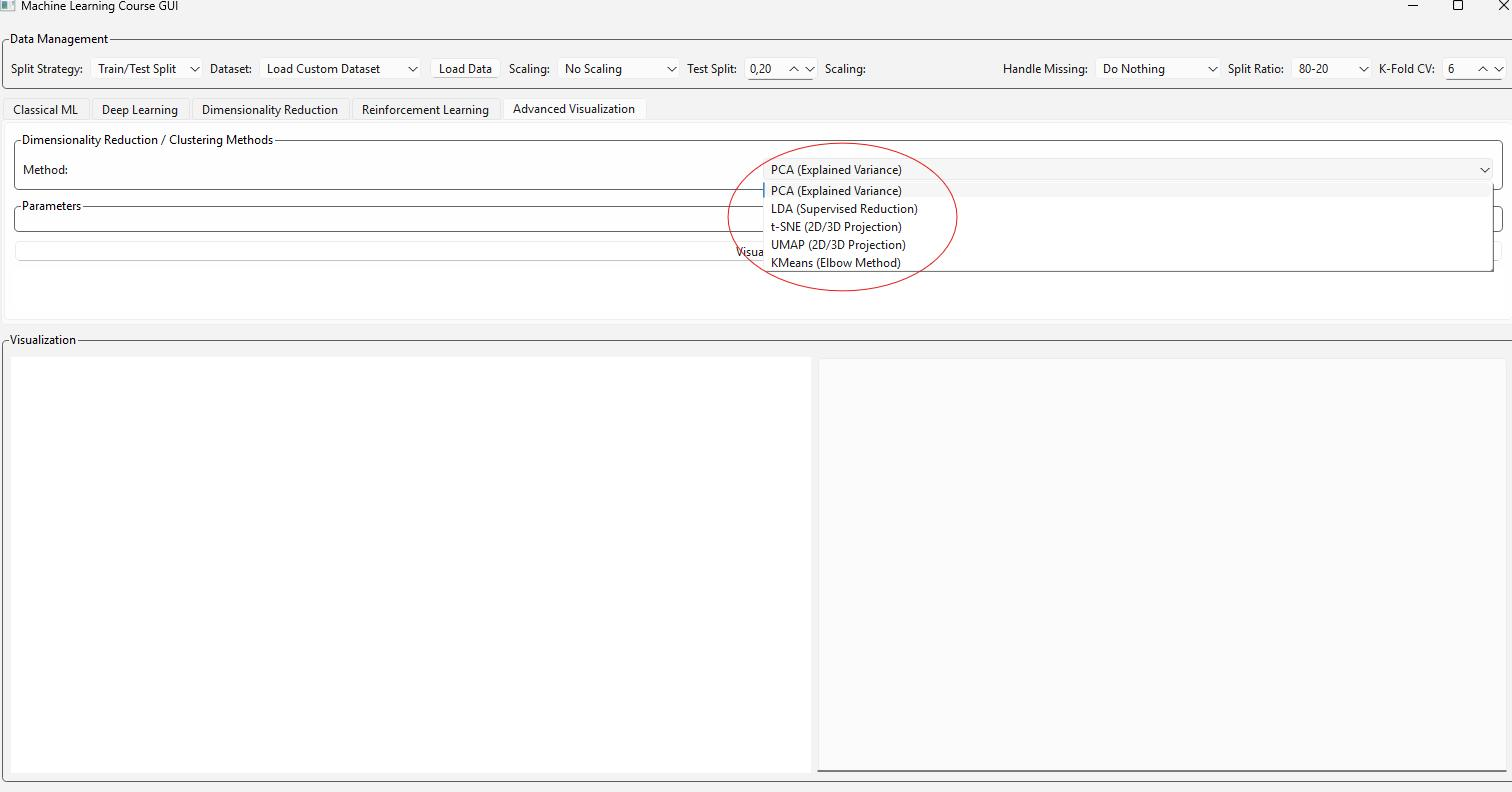
Parameters

Visualize

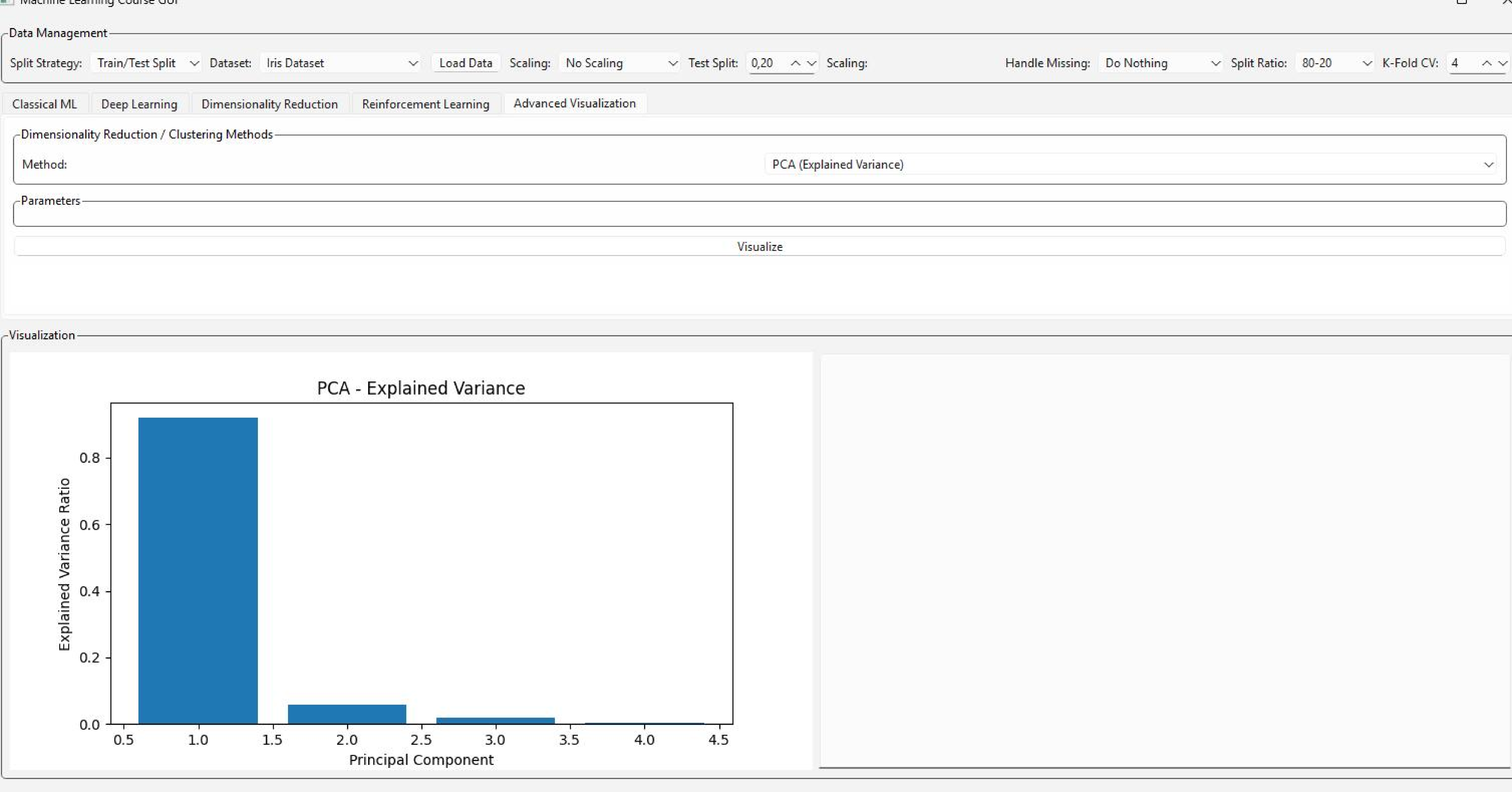
Visualization

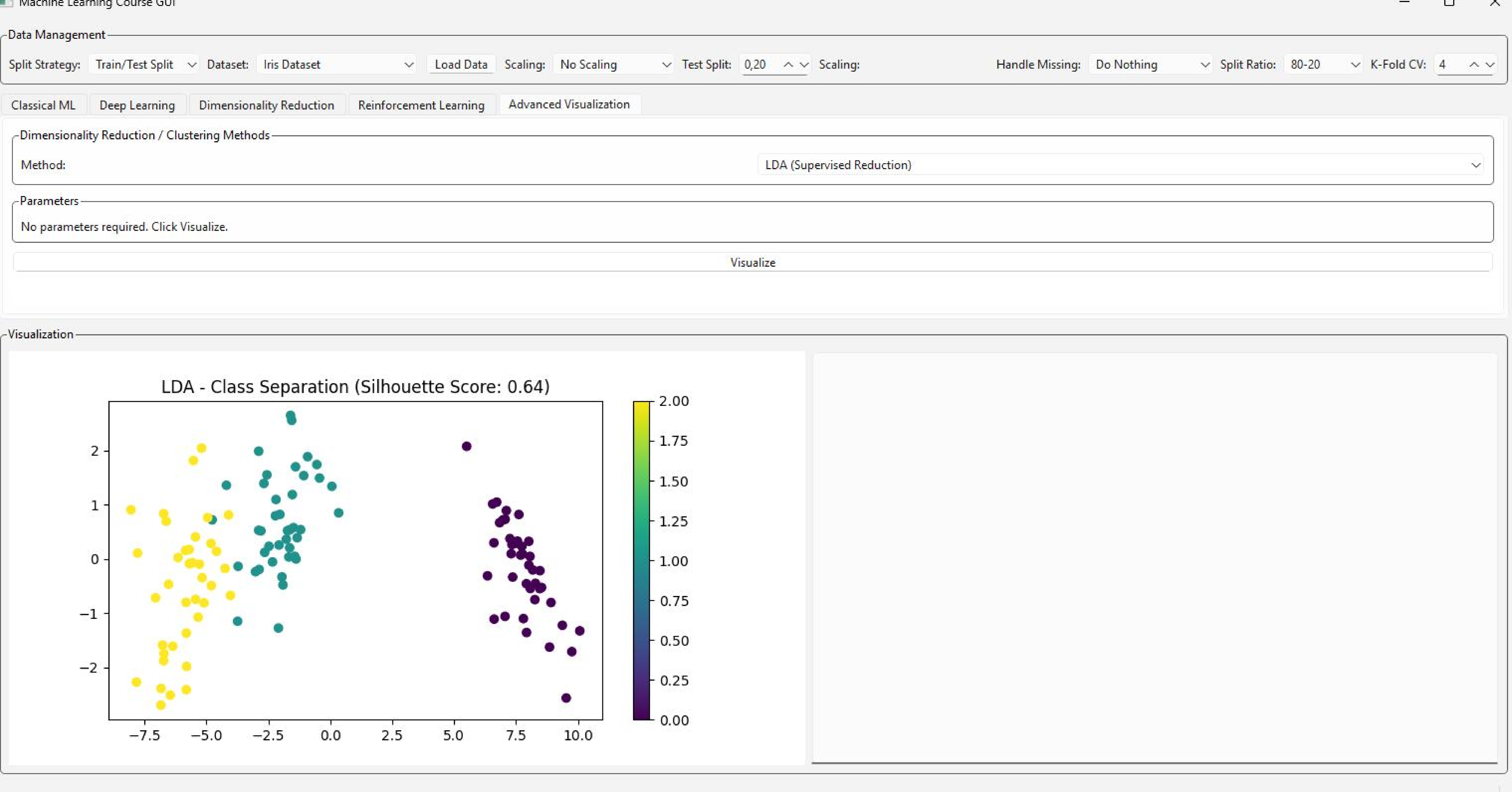

Machine Learning Course GUI
Data Management
Split Strategy: Train/Test Split
Train/Test Split
70-15-15 Split
Dataset: Load Custom Dataset
Load Data
Scaling: No Scaling
Test Split: 0,20
Scaling:
Handle Missing: Do Nothing
Split Ratio: 80-20
K-Fold CV: 5
Classical ML
Dimensionality Reduction
Reinforcement Learning
Advanced Visualization
Dimensionality Reduction / Clustering Methods
Method:
PCA (Explained Variance)
Parameters
Visualize
Visualization

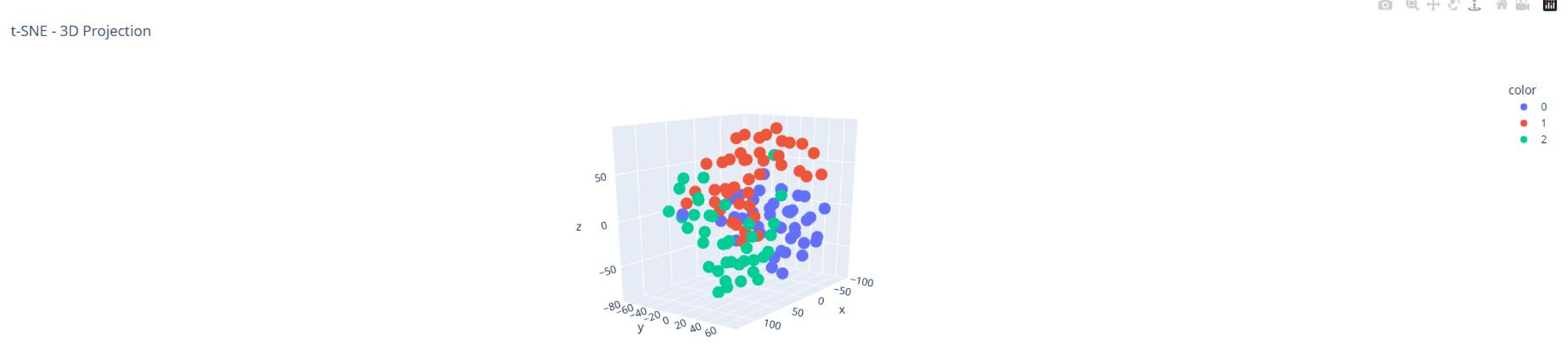

Machine Learning Course GUI
Data Management
Split Strategy: Train/Test Split
Dataset: Load Custom Dataset
Load Data
Scaling: No Scaling
Test Split: 0,20
Scaling:
Handle Missing: Do Nothing
Split Ratio: 80-20
K-Fold CV: 6
80-20
70-30
70-15-15
60-20-20
Classical ML
Deep Learning
Dimensionality Reduction
Reinforcement Learning
Advanced Visualization
Dimensionality Reduction / Clustering Methods
Method:
PCA (Explained Variance)
Parameters
Visualize
Visualization

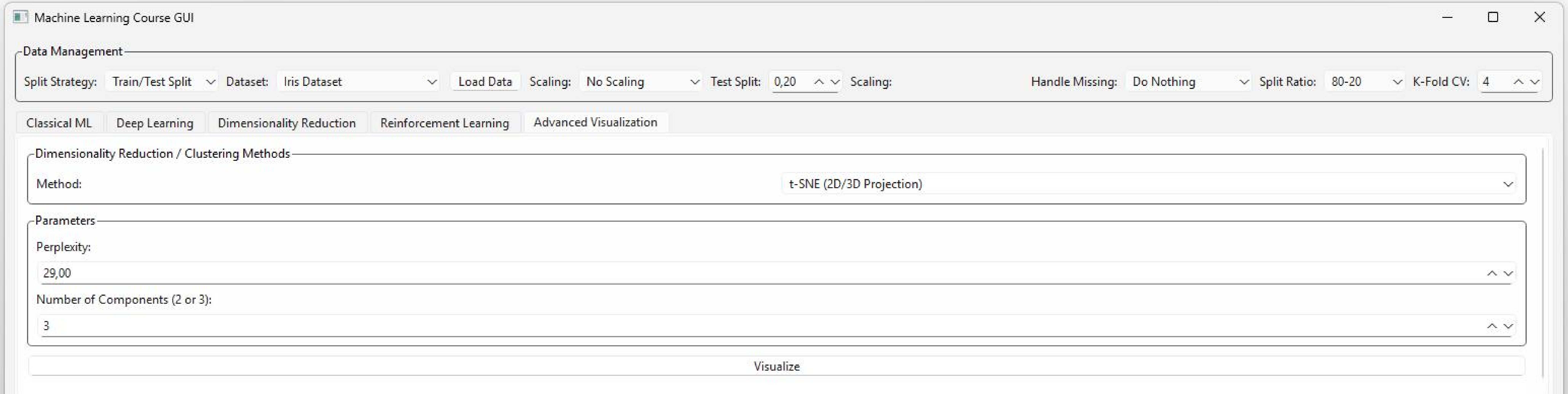


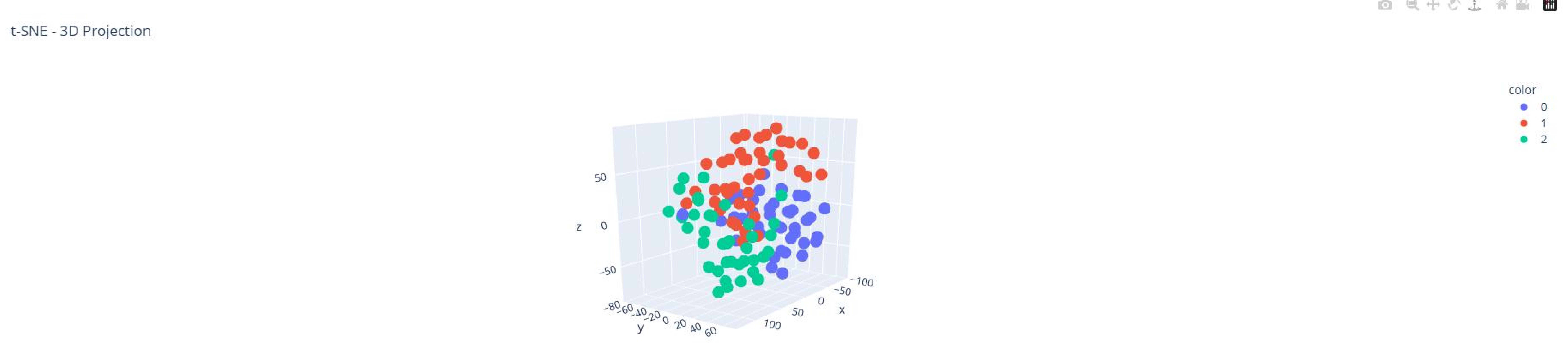
Machine Learning Course GUI
Data Management
Split Strategy: Train/Test Split
Dataset: Load Custom Dataset
Load Data
Scaling: No Scaling
Test Split: 0,20
Scaling:
Handle Missing: Do Nothing
Split Ratio: 80-20
K-Fold CV: 6
Classical ML
Deep Learning
Dimensionality Reduction
Reinforcement Learning
Advanced Visualization
Dimensionality Reduction / Clustering Methods
Method:
PCA (Explained Variance)
PCA (Explained Variance)
LDA (Supervised Reduction)
t-SNE (2D/3D Projection)
UMAP (2D/3D Projection)
KMeans (Elbow Method)
Parameters
Visualization

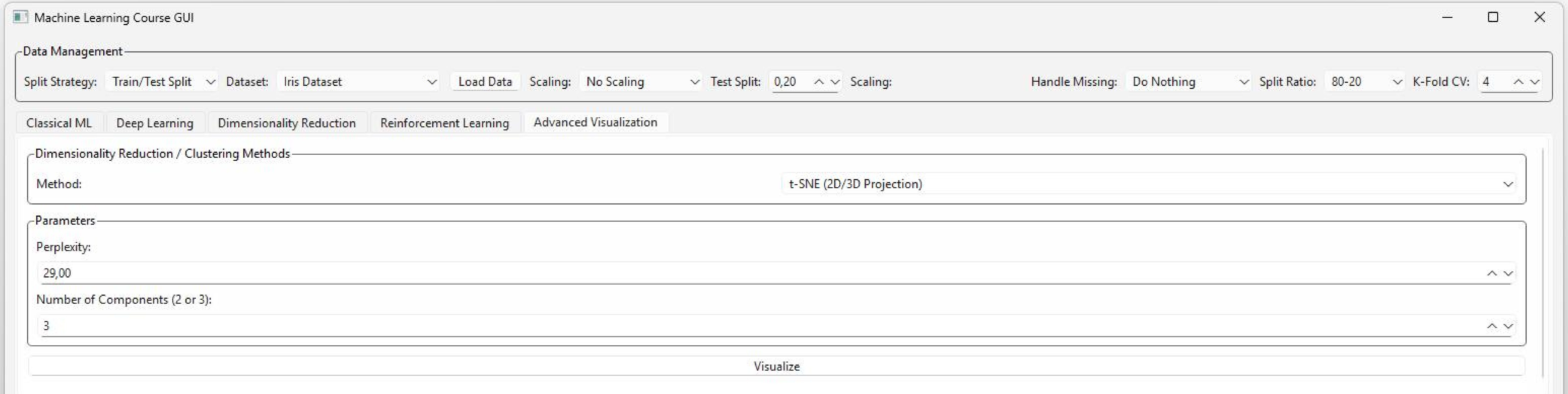

Machine Learning Course GUI
Data Management
Split Strategy: Train/Test Split
Dataset: Iris Dataset
Load Data
Scaling: No Scaling
Test Split: 0,20
Scaling:
Handle Missing: Do Nothing
Split Ratio: 80-20
K-Fold CV: 4
Classical ML
Deep Learning
Dimensionality Reduction
Reinforcement Learning
Advanced Visualization
Dimensionality Reduction / Clustering Methods
Method:
PCA (Explained Variance)
Parameters
Visualize
Visualization
PCA - Explained Variance
Explained Variance Ratio
0.0
0.2
0.4
0.6
0.8
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Principal Component


Machine Learning Course GUI
Data Management
Split Strategy: Train/Test Split
Dataset: Iris Dataset
Load Data
Scaling: No Scaling
Test Split: 0,20
Scaling:
Handle Missing: Do Nothing
Split Ratio: 80-20
K-Fold CV: 4
Classical ML
Deep Learning
Dimensionality Reduction
Reinforcement Learning
Advanced Visualization
Dimensionality Reduction / Clustering Methods
Method:
LDA (Supervised Reduction)
Parameters
No parameters required. Click Visualize.
Visualize
Visualization
LDA - Class Separation (Silhouette Score: 0.64)
2
1
0
−1
−2
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00


t-SNE - 3D Projection
color
0
1
2
50
z
0
-50
-80
-60
-40
-20
0
20
40
60
y
-100
-50
0
50
100
x


Machine Learning Course GUI
Data Management
Split Strategy:
Train/Test Split
Dataset:
Iris Dataset
Load Data
Scaling:
No Scaling
Test Split:
0,20
Scaling:
Handle Missing:
Do Nothing
Split Ratio:
80-20
K-Fold CV:
4
Classical ML
Deep Learning
Dimensionality Reduction
Reinforcement Learning
Advanced Visualization
Dimensionality Reduction / Clustering Methods
Method:
t-SNE (2D/3D Projection)
Parameters
Perplexity:
29,00
Number of Components (2 or 3):
3
Visualize


t-SNE - 3D Projection
color
0
1
2
z
50
0
−50
x
−100
−50
0
50
100
y
−80
−60
−40
−20
0
20
40
60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Method:
t-SNE (2D/3D Projection)
Parameters
Perplexity:
29,00
Number of Components (2 or 3):
3
Visualize